

Course Title: Intermediate iOS Development + Advanced SwiftUI & Architecture

Week	Topic	Lesson Breakdown (3 Hours)	Homework / Mini-Project
1	Project Setup & MVVM Architecture	● Topic: Laying the foundation with clean architecture ● Lesson Breakdown: ● Setting up the Xcode project with proper folder structure ● Introduction to MVVM with the new <code>@Observable</code> macro (iOS 16+) ● Creating the core data models: Transaction, FarmActivity, Pest, JournalEntry ● Building a FarmManager (main view model) to coordinate all features ● Live Coding: Create all data models and the main FarmViewModel class	● Set up the project with folders: Models, ViewModels, Views, Utilities ● Implement the FarmViewModel with empty arrays for each data type ● Create a main tab view with placeholders for 4 tabs (Finance, Calendar, Guide, Journal)
2	SwiftData Persistence	● Topic: Saving data permanently with SwiftData ● Lesson Breakdown: ● Converting models to SwiftData <code>@Model</code> classes ● Setting up the modelContainer and modelContext ● Understanding <code>@Query</code> for automatic UI updates ● Basic CRUD operations (Create, Read, Update, Delete) ● Live Coding: Convert all four models to SwiftData, test saving and fetching	● Complete SwiftData implementation for all models ● Add sample data on first launch ● Verify data persists after app restart
3	Programmatic Navigation & Tab Coordination	● Topic: Building a professional navigation system ● Lesson Breakdown: ● NavigationStack and NavigationPath for each tab ● Creating a NavigationCoordinator to manage navigation state ● Passing data between screens (e.g., from list to detail) ● Deep linking simulation: Opening to a specific transaction ● Live Coding: Build a navigation system where each tab has independent navigation state	● Implement a NavigationCoordinator for the Finance tab ● Create list → detail navigation for transactions ● Add "Edit" functionality with proper navigation
4	Finance Tracker Module	● Topic: Building the expense and income tracking system ● Lesson Breakdown: ● Creating the transaction list with filtering (expense/income/all) ● Building an "Add Transaction" form with category picker ● Implementing real-time calculations (total expenses, income, profit) ● Formatting currency for local users (Riel and Dollar support) ● Live Coding: Build the complete Finance tab with add/edit/delete	● Complete Finance tab with all CRUD operations ● Add a summary card showing current balance, total expenses, total income ● Implement category filtering (Seeds, Fertilizer, Labor, Tools, Sales)
5	Calendar & Reminders Module)	● Topic: Scheduling activities with local notifications ● Lesson Breakdown: ● Building a calendar view with DatePicker and custom grid ● Creating the FarmActivity model with date, type, notes ● Requesting notification permissions ● Scheduling local notifications with UNUserNotificationCenter ● Handling notification tap to open specific activity ● Live Coding: Build the Calendar tab with activity	● Complete the Calendar tab with activity CRUD ● Schedule notifications for each activity (1 day before, on the day) ● Test notifications work when app is closed

6	Pest & Disease Guide Module	- list and "Add Reminder" screen ● Topic: Building an offline reference library ● Lesson Breakdown: - Designing the Pest model with name, symptoms, treatment, image name - Creating a searchable list with searchable modifier - Implementing detail view with expandable sections - Pre-loading data from JSON file on first launch - Making it work completely offline - Live Coding: Build the Pest Guide tab with search and category filtering	● Create a JSON file with at least 10 pests/diseases ● Load this data into SwiftData on first launch ● Implement search and category tabs (Insects, Fungal, Bacterial)
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7	Daily Journal Module	● Topic: Digital notebook with rich text and weather ● Lesson Breakdown: - Building the JournalEntry model with date, content, weather, photos - Creating a timeline view of entries (reverse chronological) - Implementing weather selection (Sunny, Rainy, Cloudy, Windy) - Adding photos from library with PhotosPicker - Search and filter by date/weather - Live Coding: Build the Journal tab with entry list and add/edit screen	● Complete Journal tab with photo attachment ● Implement weather picker with SF Symbols ● Add search functionality to find entries by text
8	Dashboard & Cross-Module Integration	● Topic: Creating a unified home screen showing all data ● Lesson Breakdown: - Building a dashboard tab with summary cards - Showing recent transactions, upcoming activities, latest journal entry - Displaying total profit/loss for current month - Creating "Quick Actions" buttons for common tasks - Live Coding: Build a dashboard that pulls data from all four modules	● Create a Home tab with 4-6 summary cards ● Add navigation from each card to the relevant tab ● Show upcoming reminders for the next 7 days
9	Advanced UI & Animations	● Topic: Making the app feel polished and professional ● Lesson Breakdown: - Creating custom ViewModifiers for consistent styling - Building reusable components: FarmCard, PrimaryButton, SectionHeader - Adding subtle animations: fade-in for lists, scale for buttons - Implementing pull-to-refresh and loading states - Dark mode support and accessibility - Live Coding: Refactor the app with a design system and add animations	● Create a design system file with colors, fonts, spacing ● Build 3 reusable components and replace throughout app ● Add at least 3 animations (list appearance, button taps, transitions)
10	Data Export & Reports	● Topic: Generating useful reports for farmers ● Lesson Breakdown: - Creating monthly profit/loss reports with charts (Swift Charts) - Exporting data as CSV or PDF for sharing - Implementing UIActivityViewController for sharing reports - Building a simple PDF generator with PDFKit	● Create a monthly profit chart using Swift Charts ● Add "Share Report" button that exports CSV ● Generate a simple PDF summary of the month

		● Live Coding: Add a Reports section to Finance tab with charts and export	
11	Backup & Restore	● Topic: Helping farmers never lose their valuable data ● Lesson Breakdown: ● Understanding the importance of data backup for farmers ● Implementing export/import of all SwiftData to JSON ● Using UIDocumentPicker to save/load backup files ● Adding automatic reminder to backup weekly ● Cloud backup basics (iCloud Drive integration) ● Live Coding: Build a Backup & Restore section in Settings	● Complete backup feature that exports all data to JSON ● Implement restore that clears and imports from backup ● Test backup/restore across multiple devices
12	Final Polish & TestFlight Distribution	● Topic: Preparing for real farmer testing ● Lesson Breakdown: ● Adding app icon and launch screen (farm-themed) ● Configuring app name and bundle ID ● TestFlight setup for beta testing ● Creating a simple user guide within the app ● Planning for feedback collection ● Live Coding: Prepare and archive app for TestFlight	● Complete all remaining polish items ● Submit app to TestFlight ● Prepare final presentation/demo